Run C Report

Run C: `b5_v4_no_rag`, `b5_v4_no_ask` · Yogendra Patel · trained and scored 12–13 November 2026

> **Outcome. RAG adds no measurable value; ASK_USER clearly earns its cost.** Without retrieved policy clauses, success is identical (192/212 (90.6%) for both). Without ASK_USER, success on its 170 test cases drops from 157/170 (92.4%) to 126/170 (74.1%) ($p < 0.0001$, significant after Holm-Bonferroni) and unsafe actions rise to 8/89 (9.0%).

1. The run

Item	Value
Run by	Yogendra Patel (Kaggle `creativerebel`), branch `run-c`
Training commit	`e3d731f (13 Nov 04:38)`
Scoring commit	`aa1f572 (13 Nov 17:31)`; training merged (PR #6); scoring on branch run-c
Code	`7435346` (`locked-v4`) for training and scoring, confirmed in both kernel logs
Models	`b5_v4_no_rag`, `b5_v4_no_ask`
Scoring	NB04, model alone (no GateExp rules; the computed facts it was trained with are shown), greedy decoding, dev 236 / attacks 61 / test 212 cases, clean simulated user and approver

Training

Model	Examples	Targets	Steps	Mean loss	Time	Seed
`b5_v4_no_rag`	2,698	ALLOW 944, ASK 603, BLOCK 749, ESC 402	169	0.430	3.1 h	42
`b5_v4_no_ask`	1,952	ALLOW 725, BLOCK 927, ESC 300	122	0.407	2.7 h	42
`b5_v4_combo`	2,698	ALLOW 944, ASK 603, BLOCK 749, ESC 402	169	0.426	4.6 h	42

Last row: the final B5 for reference (same data, recipe and seed 42 unless stated).

2. Scoring results

Test pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B5 - RAG	212	192/212 (90.6%)	2/118 (1.7%)	5/110 (4.5%)	16/133 (12.0%)	189/212 (89.2%)	30/32 (93.8%)	30/39 (76.9%)	50/63 (79.4%)
B5 - ASK_USER	170	126/170 (74.1%)	8/89 (9.0%)	2/91 (2.2%)	32/108 (29.6%)	111/170 (65.3%)	25/29 (86.2%)	25/27 (92.6%)	0/52 (0.0%)
B5 alone (b5_v4_combo, seed 42)	**212**	**192/212 (90.6%)**	**3/118 (2.5%)**	**5/110 (4.5%)**	**13/133 (9.8%)**	**198/212 (93.4%)**	**39/43 (90.7%)**	**39/39 (100.0%)**	**52/63 (82.5%)**

Attacks pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B5 - RAG	61	58/61 (95.1%)	1/51 (2.0%)	2/33 (6.1%)	2/11 (18.2%)	51/61 (83.6%)	11/15 (73.3%)	11/15 (73.3%)	9/13 (69.2%)
B5 - ASK_USER	57	56/57 (98.2%)	0/47 (0.0%)	2/33 (6.1%)	1/10 (10.0%)	40/57 (70.2%)	9/13 (69.2%)	9/13 (69.2%)	0/11 (0.0%)
B5 alone (b5_v4_combo, seed 42)	**61**	**57/61 (93.4%)**	**2/51 (3.9%)**	**3/33 (9.1%)**	**2/11 (18.2%)**	**51/61 (83.6%)**	**11/15 (73.3%)**	**11/15 (73.3%)**	**9/13 (69.2%)**

Dev pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B5 - RAG	236	224/236 (94.9%)	0/112 (0.0%)	6/150 (4.0%)	9/149 (6.0%)	221/236 (93.6%)	23/23 (100.0%)	23/25 (92.0%)	54/61 (88.5%)
B5 - ASK_USER	192	141/192 (73.4%)	0/85 (0.0%)	0/123 (0.0%)	47/123 (38.2%)	128/192 (66.7%)	13/13 (100.0%)	13/16 (81.2%)	0/53 (0.0%)
B5 alone (b5_v4_combo, seed 42)	**236**	**225/236 (95.3%)**	**0/112 (0.0%)**	**9/150 (6.0%)**	**9/149 (6.0%)**	**219/236 (92.8%)**	**23/23 (100.0%)**	**23/25 (92.0%)**	**55/61 (90.2%)**

Last row in each pool: the final B5 scored the same way (model alone), the baseline every ablation is compared with.

3. Paired tests against B5

Comparison	Outcome	B5 / A	Other / B	Diff, points [95% CI]	McNemar p	Holm p
B5 vs no RAG (test)	success	192/212 (90.6%)	192/212 (90.6%)	+0.0 [-3.8, +3.8]	1.0000	1.0000
B5 vs no RAG (attacks)	success	57/61 (93.4%)	58/61 (95.1%)	-1.6 [-4.9, +0.0]	1.0000	1.0000
B5 vs no ASK_USER (test)	success	157/170 (92.4%)	126/170 (74.1%)	+18.2 [+11.8, +24.7]	**<0.0001**	<0.0001
B5 vs no ASK_USER (attacks)	success	54/57 (94.7%)	56/57 (98.2%)	-3.5 [-8.8, +0.0]	0.5000	1.0000

Paired over the same cases. Diff = B5 minus the other model (for unsafe and unneeded asks, negative means B5 is better). Exact McNemar test; bootstrap 95% interval (10,000 resamples). Green p: significant at 0.05 (after Holm-Bonferroni where the test is in the secondary family).

4. Analysis

RAG (retrieved policy clauses)

- Test: success 192/212 (90.6%) vs 192/212 (90.6%), unsafe 2/118 (1.7%) vs 3/118 (2.5%), unneeded asks equal; attacks 58/61 (95.1%) vs 57/61 (93.4%). No difference is significant (McNemar p = 1.0 on test success).
- The training fit was also the same (loss 0.430 vs 0.426) with inputs a third shorter. The model gets the policy from the structured facts (authority, provenance, sensitivity, the GateExp facts line) rather than from the clause text.
- Per the proposal ("RAG adds little: drop it from the minimal system; report Recall@k and the ablation"): RAG can be dropped from the minimal system. It still has value for explanations: the gate cites the clause it applied (clause recall@5 on test 0.785).

ASK_USER

- Success on the 170 comparable test cases: 126/170 (74.1%) vs 157/170 (92.4%) (+18.2 points [95% CI +11.8, +24.7]).
- Without the option to ask, ambiguous actions are forced into execute or block: over-blocking 32/108 (29.6%) (B5 13/133 (9.8%)) and unsafe 8/89 (9.0%) (B5 3/118 (2.5%)). Most errors are resolution choice (38).
- On attacks the model without ASK does well (56/57 (98.2%)): attacks mostly need BLOCK. The value of asking is on legitimate but under-specified requests.
- Asking costs little: B5 asks without need in 5/110 (4.5%) of cases. **ASK_USER earns its interaction cost.**

5. Records

Record	Where
Training summaries and logs	`results/data/2026-11-13_train_run_c/` (branch `run-c`)
Predictions, reports, logs	`results/data/2026-11-13_eval_run_c/` (branch `run-c`)
Statistics over all runs	`results/data/2026-11-14_analysis/`